

Learning Guide: Architectural Design Ideation and Visualisation with Generative AI

Team 11

Why, What and How

Why (impact & scalability on practicality)

Consultants are often challenged by demanding requirements with tight deadlines, rapid technological advancements requiring quick adaptation, and the need to close skills gaps and enhance competency.

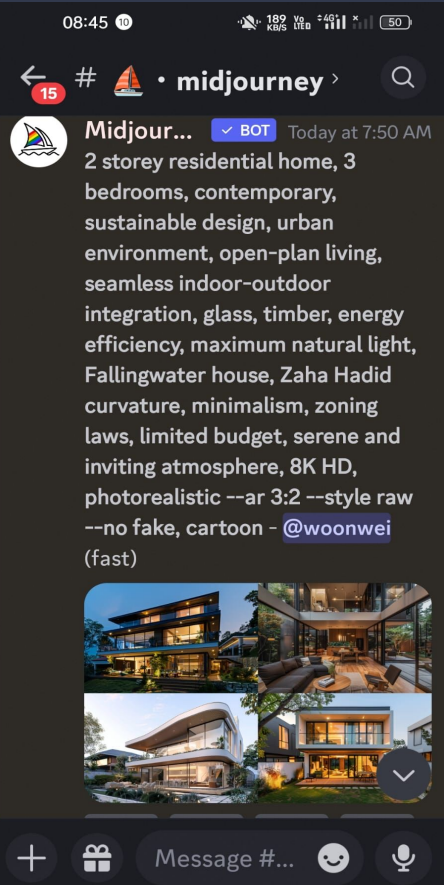
What (innovation & integration to engage GenAI)

This project aims to address the challenge of creating user-responsive architectural design ideations and visualisations by exploring Generative AI tools and providing accessible learning resources shared among the community of practice.

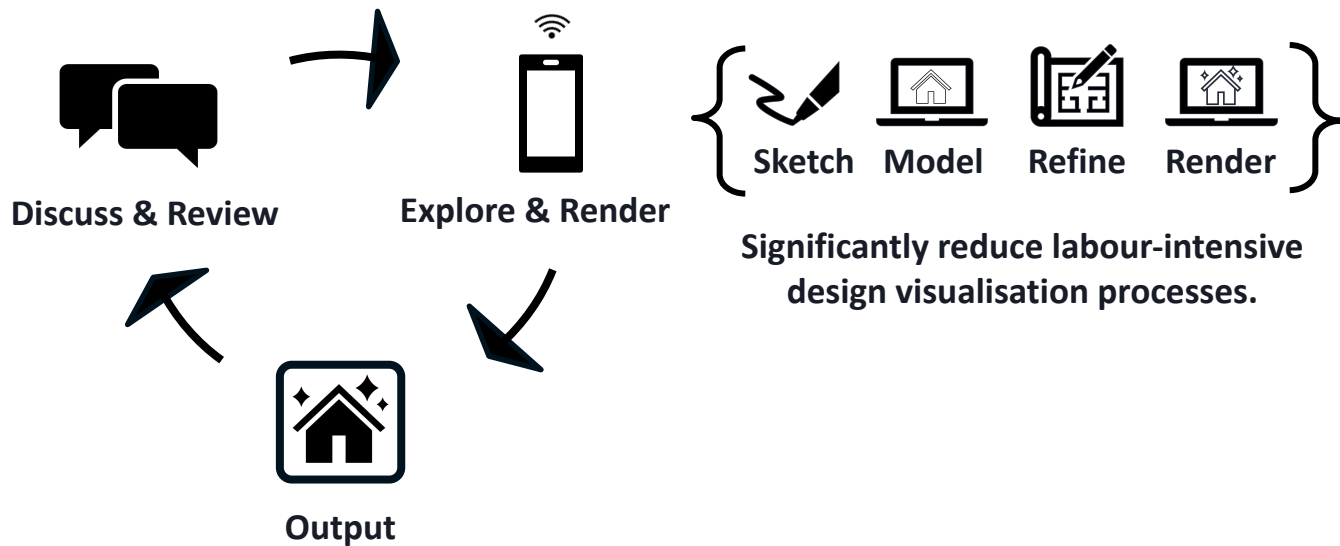
How (Documentation & Implementation through open sharing)

The solutions involved online and local Generative AI tools using custom datasets, gathered practitioner feedback on time and effort savings, and used techniques with massing models to enhance design accuracy and meet development needs. A Guide was developed and shared openly.

Midjourney: Architectural Design Ideation and Visualisation Approach



The left image showcased Midjourney's user interface on a Discord channel, enabling consultants to create design visuals anytime and anywhere using a simple prompt. This method significantly reduces the traditional labour-intensive upfront design visualisation processes.



Midjourney: Process of Ideating and Visualizing Architectural Designs

March 5, 2024



Midjourney Bot ✓ BOT Today at 7:50 AM

2 storey residential home, 3 bedrooms, contemporary, sustainable design, urban environment, open-plan living, seamless indoor-outdoor integration, glass, timber, energy efficiency, maximum natural light, Fallingwater house, Zaha Hadid curvature, minimalism, zoning laws, limited budget, serene and inviting atmosphere, 8K HD, photorealistic --ar 3:2 --style raw --no fake, cartoon - @woonwei (fast)

prompt



Variations



Upscaled Image



Outputs

U1

U2

U3

U4



V1

V2

V3

V4

Midjourney: Quick Iterations of Architectural Design



✓ BOT Midjourney Bot 2 storey residential home, 3 bedrooms, contemporary, sustainable design, urban environment, open-plan living, seamless indoor-outdoor integrat

Midjourney Bot ✓ BOT Today at 7:51 AM

2 storey residential home, 3 bedrooms, contemporary, sustainable design, urban environment, open-plan living, seamless indoor-outdoor integration, glass, timber, energy efficiency, maximum natural light, Fallingwater house, Zaha Hadid curvature, minimalism, zoning laws, limited budget, serene and inviting atmosphere, 8K HD, photorealistic --ar 3:2 --style raw --no fake, cartoon - Variations (Strong) by [@woonwei](#) (fast)



U1

U2

U3

U4



V1

V2

V3

V4

Midjourney: Quick Iterations of Interior Design



✓ BOT Midjourney Bot 2 storey residential home, 3 bedrooms, contemporary, sustainable design, urban environment, open-plan living, seamless indoor-outdoor integrat

Midjourney Bot ✓ BOT Today at 7:51 AM

2 storey residential home, 3 bedrooms, contemporary, sustainable design, urban environment, open-plan living, seamless indoor-outdoor integration, glass, timber, energy efficiency, maximum natural light, Fallingwater house, Zaha Hadid curvature, minimalism, zoning laws, limited budget, serene and inviting atmosphere, 8K HD, photorealistic --ar 3:2 --style raw --no fake, cartoon - Variations (Strong) by @woonwei (fast)



U1

U2

U3

U4



V1

V2

V3

V4

Midjourney: Quick Refinement of Interior Design



✓ BOT Midjourney Bot 2 storey residential home, 3 bedrooms, contemporary, sustainable design, urban environment, open-plan living, seamless indoor-outdoor integrat

Midjourney Bot ✓ BOT Today at 8:00 AM

2 storey residential home, 3 bedrooms, contemporary, sustainable design, urban environment, open-plan living, seamless indoor-outdoor integration, glass, timber, energy efficiency, maximum natural light, Fallingwater house, Zaha Hadid curvature, minimalism, zoning laws, limited budget, serene and inviting atmosphere, 8K HD, photorealistic --ar 3:2 --style raw --no fake, cartoon - Zoom Out by @woonwei (fast)



U1

U2

U3

U4



V1

V2

V3

V4

Midjourney: Quick Upscaling of Interior Design Visuals



✓ BOT Midjourney Bot 2 storey residential home, 3 bedrooms, contemporary, sustainable design, urban environment, open-plan living, seamless indoor-outdoor integrat

Midjourney Bot ✓ BOT Today at 8:18 AM

2 storey residential home, 3 bedrooms, contemporary, sustainable design, urban environment, open-plan living, seamless indoor-outdoor integration, glass, timber, energy efficiency, maximum natural light, Fallingwater house, Zaha Hadid curvature, minimalism, zoning laws, limited budget, serene and inviting atmosphere, 8K HD, photorealistic --ar 3:2 --style raw --no fake, cartoon --v 6.0 - Upscaled (Subtle) by @woonwei (fast)



Redo Upscale (Subtle)

Redo Upscale (Creative)

Vary (Subtle)

Vary (Strong)

Midjourney: Prompt Engineering Structure and Techniques

An Excel spreadsheet was created to showcase the Prompt Engineering Structure and Techniques for generating high-quality images, such as the use of image URLs to guide image generation, text structure to describe content, and parameters to define the setup needed in the Midjourney tool.

A	B	C	D	E	F
No	Component	Item	Keywords and Description	Example for Midjourney Prompt	Midjourney Parameters
1	Image URL (Weighting)	Image URL	External images to guide generation, with influence weights.	"URL Image 1 :: 3, URL Image 2 :: 1,"	URL Image 1 :: 3 URL Image 2 :: 1
2	Text	Subject & Medium	Type of Photography, Building typology	Architecture, Exterior	
3		Environment	indoor, outdoor	Urban, Cityscape, Countryside	
4		Architectural Styles	Focuses on architectural features in the image.	Gothic elements, Flying buttresses, MVDVR	
5		Architects' Names	notable architects or firms	Zaha Hadid, Peter Eisenman, etc.	
6		Design Elements	Facade, Details	Glass Facade, with Steel Renderings	
7		Surrounding	Detail building surrounding context	under a clear blue sky	
8		Light	The quality and source of light in the image.	Dynamic, Twilight, Natural Light,	
9		Rendering Style & Color	Image composition, style & color	MIR rendering style, vibrant, bright, pastel, monochrome	
10		Mood	The emotional tone or feeling of the image.	Serene, Vibrant	
11		Point of View	The angle or perspective from which the image is taken.	Bird's-eye view, First-person 2-point perspective view, isometric	
12	Parameters	Version, Aspect Ratio, Style, Stylize, Style Reference	Technical settings for customization.	"Using niji for a more focused detail, aspect ratio --ar 16:9,"	"--v 6, --s 500, --stylize 700, --sref <image url>"
URL Image 1 :: 3, URL Image 2 :: 1 -- Modern residential architecture, influenced by Brutalism. Inspired by Zaha Hadid's fluid forms. Incorporating expansive glass facades and intricate steelwork, preferably in a MIR-like rendering style. Aiming for a serene yet vibrant atmosphere, set in an urban landscape amidst a lush green park, under a clear blue sky. Capturing the dynamic twilight ambiance, from a bird's-eye view. --v 6 --ar 16:9 --s 500 --stylize 700 --sref <image url>					
a 8 level retail indoor perspective photograph, interconnectivity, floors, varied stores, storefronts, trees, Feature escalators and elevators, glass, timber --seed 3953343062 --sref https://s.mj.run/4YSetrsalDl --style raw					
https://s.mj.run/4YSetrsalDl					
Prompt Length and Details					
Prompts can be simple. A single word or emoji will work. However, short prompts rely on Midjourney's default style, allowing it to fill in any unspecified details creatively. Include any element that is important to you in your prompt. Fewer details means more variety but less control.					
Try to be clear about any context or details that are important to you. Think about:					
• Subject: person, animal, character, location, object • Medium: photo, painting, illustration, sculpture, doodle, tapestry • Environment: indoors, outdoors, on the moon, underwater, in the city • Lighting: soft, ambient, overcast, neon, studio lights • Color: vibrant, muted, bright, monochromatic, colorful, black and white, pastel • Mood: sedate, calm, raucous, energetic • Composition: portrait, headshot, closeup, birds-eye view					

Midjourney: An Overview of Architecture Styles

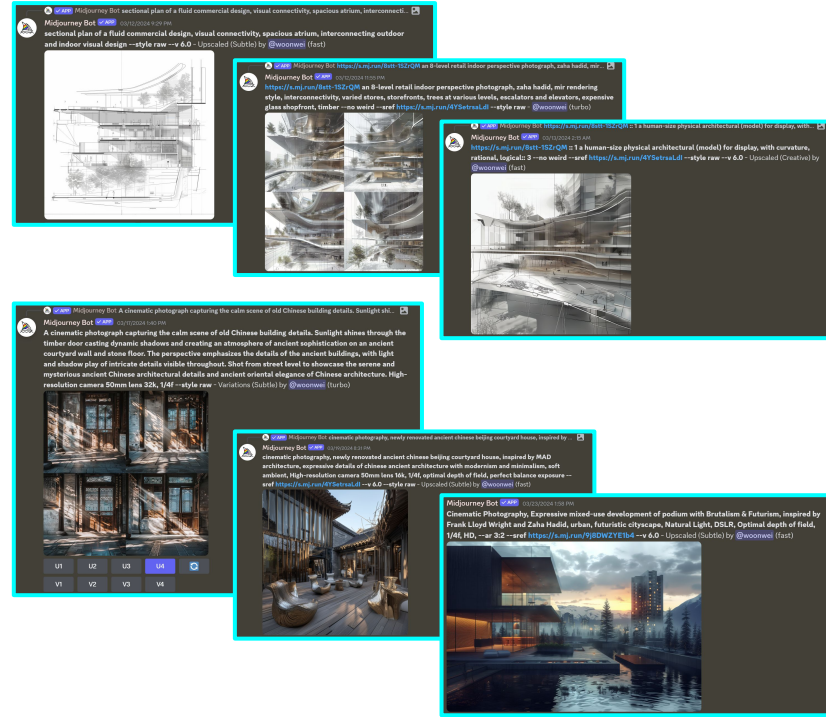
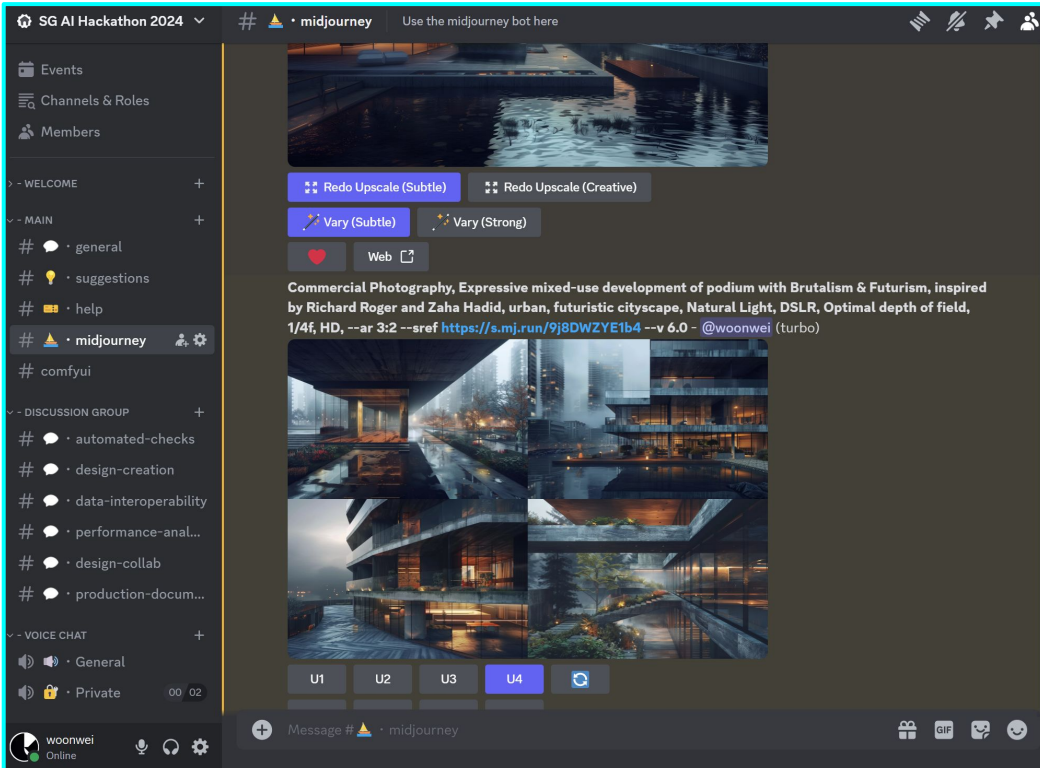
A table was created with a list of 62 architectural styles along with their corresponding time periods, notable architects, building typologies, and key projects for easy reference.

	A	B	C	D	E	F
1	No	Architectural Style	Time Period	Notable Architects	Building Typology	Notable Projects
2	1	Ancient Egyptian	2600-330 BC			
3	2	Ancient Sumerian				
4	3	Ancient Greek				
5	4	Ancient Roman				
6	5	Byzantine				
7	6	Romanesque				
8	7	Persian				
9	8	Pre-Columbian				
10	9	Mesoamerican				
11	10	Indian				
12	11	Chinese				
13	12	Japanese				
14	13	Islamic				
15	14	Norse				
16	15	African Traditional				
17	16	Stave Church Style				
18	17	Gothic				
19	18	Renaissance				
20	19	Baroque				
21	20	Rococo				
22	21	Neoclassicism				
23	22	Beaux-Arts				
24	23	Russian Revival				
25	24	Art & Crafts Movement				
26	25	Art Nouveau				
27	26	Prairie Style				
28	26	Craftsman/Bungalow				
29	27	Art Deco				
30	28	Organic Architecture				
31	29	Futurism				
32	30	Futurist Architecture				
33	31	Expressionism				
34	32	Expressionist Modernism				
35	33	Pueblo Revival				
36	34	Constructivism				
37	35	Bauhaus				
38	36	International Style				
39	37	Streamline Moderne				
40	38	Stalinist Architecture				
41	39	Scandinavian Modernism				
42	40	Mid-Century Modern				
43	41	Structuralism				
44	42	Brutalism				
45	43	Metabolism				
46	44	Minimalism				
47	45	Post-Structuralism				
48	46	Postmodernism				
49	47	High-Tech				
50	48	Contemporary				
51	49	Critical Regionalism				
52	50	New Classical				
53	51	Deconstructivism				
54	52	Neo-Traditional				
55	53	Critical Regionalism				
56	54	Neo-Futurism				
57	55	Blobitecture				
58	56	Adaptive Reuse				
59	57	Sustainable Architecture				
60	58	Parametricism				
61	59	Digital Architecture				
62	60	Modular Architecture				
63	61	Neo-Brutalism				
64	62	Smart Architecture				
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Download Link: <https://forms.gle/QpgbwWsv8cK1zsmk9>

Midjourney: Discord Channel Sharing for Community Engagement

This Discord channel showcases the journey from the very beginning of learning Midjourney, progressing to the use of structured prompt engineering techniques.



Discord Channel: <https://discord.gg/mpVfAGfX>

ComfyUI: Steps to create text files based on prepared images

Generative AI may accelerate personalised designs by using own datasets for own model training.

25 Images were prepared for the model training. Screenshots below shows the steps using ComfyUI to create text files that is related to the images using WD 14 Tagger custom nodes.

1. Double Right Click “run_nvidia_gpu.bat” to open ComfyUI

2. Load ‘json’ file

3. Update image folder, e.g. “C:\database\5_brutalism”

4. Click “Queue Prompt” to let the AI Engine to create text files for provided images

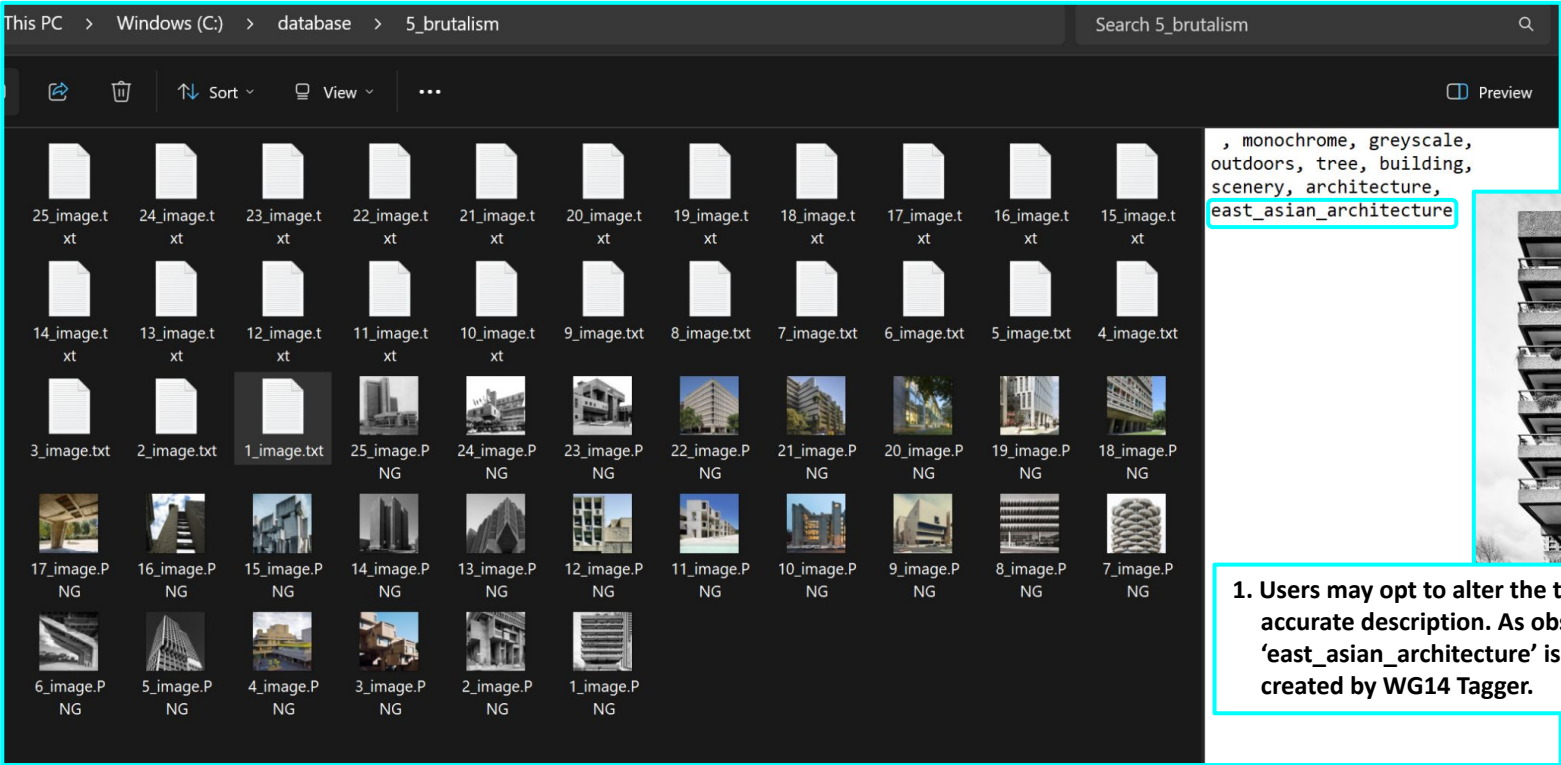
json file loaded in ComfyUI

Provided Images stored under “C:\database\5_brutalism”

Link (json files): <https://drive.google.com/drive/folders/1gSCx6bxPZuZOIAdBaS1wiLtYKGKZRzcn?usp=sharing>

ComfyUI: Created text files based on prepared images

Screenshot below showed that the respective text files were created referenced to the prepared images, for the LoRA model training.



1. Users may opt to alter the text file for more accurate description. As observed, 'east_asian_architecture' is one of the key words created by WG14 Tagger.

ComfyUI: Steps to create brutalism.safetensors file

Screenshot below showed the use of a Lora Training node in ComfyUI, loaded in a web browser user interface, to create the brutalism.safetensors file based on images & text saved under “C:\database\5_brutalism”. Users are required to set data_path as “C:\database”.

2. Right click “Queue Prompt” to run the training

Queue Prompt

Extra options

Queue Front

View Queue

View history

Save

Load

Refresh

Clipspace

Clear

Load Default

Manager

Share

1. Users are required to set the data_path as “C:\database”

data_path

C:\database

Custom node

Lora Training in ComfyUI

ckpt_name

v1-5-pruned.safetensors

batch_size

1

max_train_epochs

10

save_every_n_epochs

10

output_name

brutalism

clip_skip

2

output_dir

models\loras

Training Process

```
running training / 学習開始
num train images / repeats / 学習画像の数×繰り返し回数: 125
num reg_images / 正則化画像の数: 0
num batches per epoch / 1epochのバッチ数: 125
num epochs / epoch数: 10
batch size per device / バッチサイズ: 1
gradient accumulation steps / 勾配を合計するステップ数 = 1
total optimization steps / 学習ステップ数: 1250

steps: 0% | 0/1250 [00:00<?, ?it/s]
epoch 1/10
steps: 10% | 125/1250 [01:15<11:18, 1.66it/s, avr_loss=0.104]
epoch 2/10
steps: 20% | 250/1250 [01:57<07:49, 2.13it/s, avr_loss=0.12]
epoch 3/10
steps: 30% | 375/1250 [02:39<06:11, 2.35it/s, avr_loss=0.132]
epoch 4/10
steps: 40% | 500/1250 [03:21<05:01, 2.48it/s, avr_loss=0.139]
epoch 5/10
steps: 50% | 625/1250 [04:03<04:03, 2.57it/s, avr_loss=0.123]
epoch 6/10
steps: 60% | 750/1250 [04:44<03:09, 2.63it/s, avr_loss=0.109]
epoch 7/10
steps: 70% | 875/1250 [05:26<02:19, 2.68it/s, avr_loss=0.129]
epoch 8/10
steps: 80% | 1000/1250 [06:08<01:32, 2.72it/s, avr_loss=0.127]
epoch 9/10
steps: 90% | 1125/1250 [06:49<00:45, 2.75it/s, avr_loss=0.11]
epoch 10/10
steps: 100% | 1250/1250 [07:31<00:00, 2.77it/s, avr_loss=0.129]
saving checkpoint: models\loras\brutalism.safetensors
model saved.
Train finished
Prompt executed in 480.49 seconds
```

Relocation of file

4. Relocate the brutalism.safetensors to “C:\..\ComfyUI_windows_portable\ComfyUI\models\loras”, which is the default folder set for ComfyUI to read models

3. brutalism.safetensors file is created, saved under “C:\..\ComfyUI_windows_portable\models\loras”, after the training is complete

Created safetensors location

brutalism.safetensors

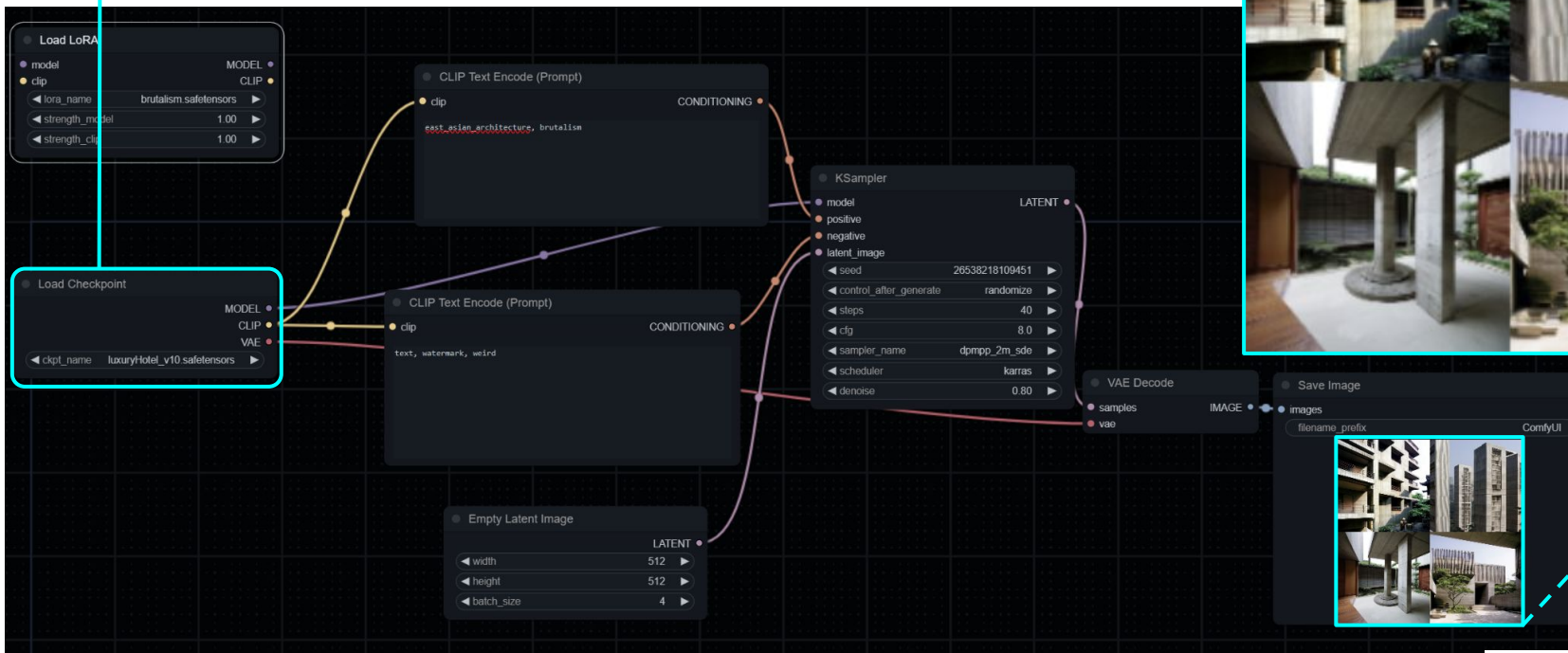
18/4/2024 2:59 pm

SAFETENSORS File

36,978 KB

ComfyUI: Architectural Design Ideation and Visualisation

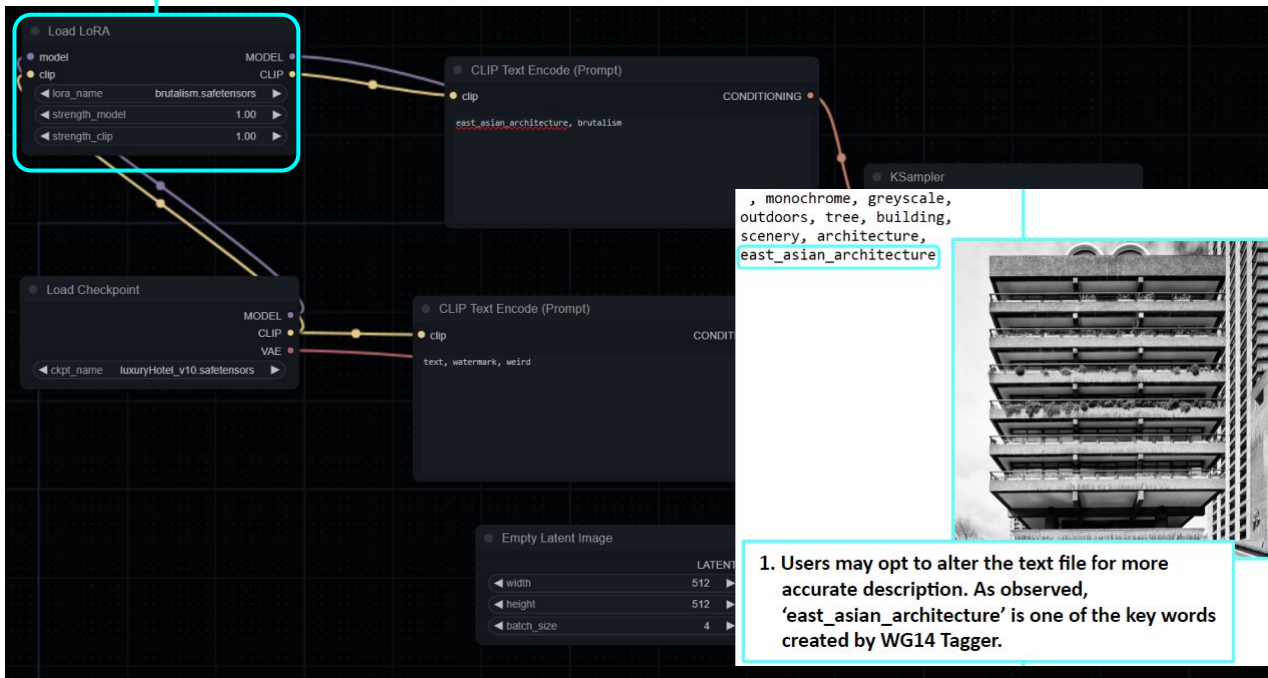
Screenshot below showed the use of a **downloaded checkpoint model** named **luxury_Hotel.safetensors** file as the datasets to render the outcome base on the simple prompt of **east_asian_architecture** and **brutalism**



Link (json files): <https://drive.google.com/drive/folders/1gSCx6bxPZuZOIAdBaS1wiLtYKGKZRzcN?usp=sharing>

ComfyUI: Architectural Design Ideation and Visualisation

Screenshot below showed the use of a downloaded checkpoint model named `luxury_Hotel.safetensors` file together **with self-trained Lora file** to render the outcome base on the simple prompt of `east_asian_architecture` and `brutalism`. The scale and texture is affected by the key word of `east_asian_architecture`.



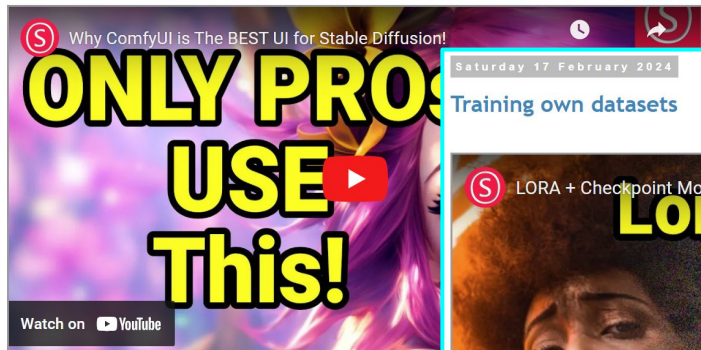
1. Users may opt to alter the text file for more accurate description. As observed, 'east_asian_architecture' is one of the key words created by WG14 Tagger.

ComfyUI: Open Sharing of curated resources for Community Engagement

Learn from the curated resources for ComfyUI installation, Model training using own datasets, and more at <https://www.integrationass.space/blog>.

Wednesday 24 January 2024

Stable Diffusion ComfyUI



More YouTube Tutorials:

- <https://www.youtube.com/watch?v=VM9nsnuoqBc>

Downloading ComfyUI

1. Download ComfyUI Zip file:

<https://github.com/comfyanonymous/ComfyUI/releases/download/latest/cpu.7z>

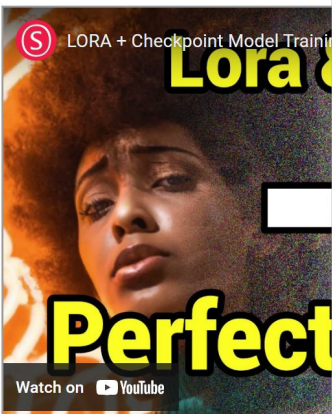
2. Unzip the downloaded Zip file, right click the file -> properties -> unblock

3. After unzipping, go to "ComfyUI_windows_portable" folder and double click screen below.

4. Click "Load Default" and close the window.

Saturday 17 February 2024

Training own datasets



at February 17, 2024 No comments:

Labels: [Blog](#), [Stable Diffusion](#)

More YouTube Tutorials:

- <https://www.youtube.com/watch?v=VM9nsnuoqBc>

Downloading ComfyUI

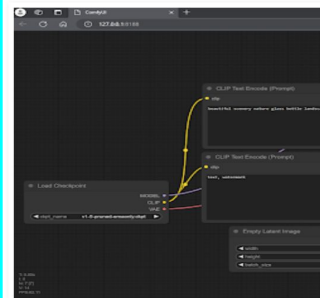
1. Download ComfyUI Zip file:

<https://github.com/comfyanonymous/ComfyUI/releases/download/latest/cpu.7z>

2. Unzip the downloaded Zip file, right click the file -> properties -> unblock

3. After unzipping, go to "ComfyUI_windows_portable" folder and double click screen below.

4. Click "Load Default" and close the window.



5. Locate "ComfyUI\com_nodes" directory by searching your folder and copy file path as text

6. Command Prompt > cd "your copied file path"

7. Command Prompt > paste "git clone https://github.com/ltldrdata/ComfyUI-Manager.git" to install ComfyUI-Manager

Link: <https://github.com/ltldrdata/ComfyUI-Manager?tab=readme-ov-file> (5-7 steps as method 1 above)

Open Sharing of Knowledge and Initiatives in Computational BIM and Digital Technology within the Community of Practice.

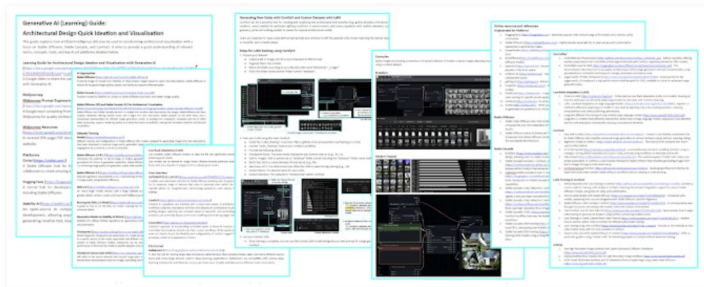
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Friday 19 April 2024

Generative AI (Learning) Guide

I have developed a Generative AI (Learning) Guide available at URL here:

https://docs.google.com/document/d/1bbSOV8KU564R3r7IMLo8D__DJjQ0j0xemPErpdhCG_1/edit#heading=h.po0cc9dogcb9



Open sharing with Community of Practice: <https://www.integrations.space/blog>

What Next

Explore different AI tools and GenAI model training

Compile and assess various AI tools for industry awareness, and broaden model training to include diverse building types and tropical designs for architectural projects

Extend GenAI to engineering and tender drawing production through BIM integration

Explore linking GenAI's with BIM software to automate drawing generation using predefined BIM modules for rapid detailing and design refinement.

Explore topic-focused hands-on activities

Engage industry partners in hands-on AI / Gen AI activities and trials to build digital competency and foster community growth.

Ideate. Innovate. Integrate

Thank you :)

- Ideate with insights to unleash creativity
- Innovate with actions to enhance productivity
- Integrate with impact to drive transformation